Chapter 1

INTRODUCTION

1.1 Preamble:

Agriculture is the backbone of many economies, as we all know agriculture is a non-technical industry where technology can help with growth and improvement. Agricultural technology needs to be implemented quickly and in a growing number of places. This industry has employed almost 60% of the population of our country. On the other hand, it accounts for only 18 percent of India's annual GDP. The disparity arises from A lack of research and innovative technology. The proper selection of crops is crucial for maximizing yield and sustainability.

However, farmers often face challenges in determining the most suitable crops for their land due to varying factors such as soil type, weather conditions, and market demand. Traditional methods of crop selection rely heavily on personal experience, which can lead to inconsistent results and inefficiencies. Farmers using traditional methods for identifying which crops to grow often end up while growing inefficient crops, while using technologies can help in growing much more efficient crops. In recent years, advancements in machine learning have opened new opportunities to revolutionize agriculture.

A crop recommendation system powered by machine learning analyses vast amounts of data, including soil properties, climate, and historical crop performance, to suggest the most appropriate crops for a given region. This not only improves crop productivity but also promotes resource efficient farming, helping farmers make informed decisions.

The Crop Recommendation Project is an initiative that aims to help farmers select the most suitable crops for cultivation based on various environmental, soil, and economic factors. This project leverages advanced data science techniques, particularly **Machine Learning (ML)** algorithms, to analyse complex datasets, providing actionable insights to maximize agricultural productivity.

Machine learning (ML) has played an increasingly significant role in agriculture, revolutionizing traditional practices and addressing challenges such as population growth, climate change, and resource scarcity. Early use of data analysis in agriculture began with precision farming, which relied on data from satellite imagery and sensors. While not strictly machine learning, this laid the groundwork for advanced algorithms. Researchers started using statistical models to optimize crop yields, irrigation, and fertilizer use. ML algorithms started being applied to classify crops, predict yields, and detect plant diseases using data from soil sensors, weather stations, and remote sensing technologies.

Early applications focused on supervised learning for specific tasks like identifying pests and diseases based on image recognition. The proliferation of drones, IoT devices, and mobile apps revolutionized data collection, providing massive datasets for training ML models. ML enabled real-time analysis of weather patterns, soil quality, and crop health, helping farmers make timely decisions. Innovations like autonomous tractors and robotics used ML for navigation and task execution. ML contributes to sustainable farming by minimizing resource use (water, fertilizer, and pesticides) and reducing environmental impact.

It empowers small-scale farmers with accessible technologies, improving livelihoods in developing regions. The integration of machine learning in agriculture is a critical development in ensuring sustainable and efficient food production, adapting to changing environmental conditions, and addressing global challenges.

The primary goal of this project is to offer farmers data-driven recommendations, allowing them to make informed decisions on which crops will thrive in their particular region or conditions. By considering factors such as soil properties, climate, water availability, and market demand, the crop recommendation system can guide farmers toward high-yield and cost-effective farming practices, ultimately contributing to food security and sustainable agricultural practices. Our main goal of this project is to help farmers grow better crops by using technology.

1.2 Key Factors in Crop Recommendation:

Soil Type and Quality: Soil is a key factor influencing crop growth. Factors like soil pH, texture, fertility, and nutrient content determine the suitability of different crops.

Weather and Climate: Climate conditions such as temperature, humidity, rainfall, and sunlight play a major role in crop growth. Machine learning models can predict how these variables affect different crops and suggest the most suitable ones for a particular season.

Water Availability: The availability of water (irrigation) is another significant factor. Crops vary in their water needs, and water scarcity or abundance can greatly affect crop selection.

Pests and Diseases: Historical data about pests and diseases in specific regions can help recommend crops that are more resistant or less prone to these issues.

Market Demand and Economic Factors: Crop recommendation systems can also take into account economic factors like market demand and crop price trends to recommend profitable crops.

1.3 Steps in Developing a Crop Recommendation System Using ML algorithms:

Data Collection:

Historical data on crop yields, weather patterns, soil properties, pest occurrences, etc.

External data sources such as satellite imagery or remote sensing data for climate and soil analysis.

Data Preprocessing:

Cleaning the dataset by handling missing values, removing irrelevant features, and normalizing data.

Feature engineering to create new features that might improve model performance (e.g., converting weather data into seasonality trends).

Choosing the Machine Learning Algorithm: Several algorithms can be applied to crop recommendation, depending on the type of data and the goal of the model:

Supervised Learning:

Regression algorithms like Linear Regression or Random Forest Regressor can predict continuous values, such as crop yield.

Classification algorithms like Logistic Regression, Support Vector Machines (SVM), or Decision Trees can classify areas based on the suitability of different crops.

Unsupervised Learning:

Clustering algorithms like K-means can be used to group regions with similar environmental factors and recommend similar crops for these regions.

Reinforcement Learning:

This approach can be used to continuously improve crop recommendations by learning from past decisions, yielding higher efficiency and accuracy over time.

Model Training and Evaluation:

Train the selected model(s) using the training dataset and evaluate their performance on a separate validation dataset.

Evaluation metrics like accuracy, precision, recall, or Root Mean Squared Error (RMSE) can be used depending on whether the problem is a classification or regression task.

Deployment and Feedback:

Deploy the model to a real-world environment, where farmers can input their local conditions (e.g., soil type, weather data) to get crop recommendation. Incorporate feedback to continually improve the model.

1.4 Popular Machine Learning Algorithms Used in Crop Recommendation:

Decision Trees: A simple yet effective algorithm for classification and regression tasks. Decision trees are easy to interpret and can provide insights into the decision-making process behind crop recommendations.

Random Forests: An ensemble method based on decision trees that provides more accurate and robust predictions by averaging over many decision trees to avoid overfitting.

Support Vector Machines (SVM): Often used for classification problems, SVM can help determine the boundaries between suitable and unsuitable crops based on environmental factors.

K-Nearest Neighbours (KNN): This algorithm classifies crops based on the similarity between the environmental features of one region and another.

Neural Networks: Deep learning models, such as Artificial Neural Networks (ANN), can handle large and complex datasets, learning intricate relationships between various factors affecting crop growth.

K-Means Clustering: An unsupervised learning technique that can group similar regions, based on their environmental and soil conditions, and recommend similar crops for those.

Gradient Boosting Algorithms: Techniques like XGBoost or LightGBM are very powerful for predicting crop yields or selecting crops, especially in datasets with many variables.

1.5 Problem statement:

Identifying of crops to grow on particular area based on soils pH and temperature using machine learning algorithm.

Agriculture plays a crucial role in the global economy, providing food, raw materials, and employment to millions of people. However, farmers often face challenges in selecting the most suitable crops for their specific region due to factors such as unpredictable weather conditions, soil quality, water availability, pest attacks, and market demand. Incorrect crop choices can lead to poor yields, resource wastage, and financial losses. Farmers in various regions struggle with making data driven decisions about which crops to grow.

Traditional methods of crop selection, based on experience and trial-and-error, may not always yield optimal results, particularly as environmental factors and market conditions change. There is a need for a system that can provide personalized crop recommendations based on a comprehensive analysis of environmental, soil, and economic data. This recommendation system should help farmers optimize crop yield, reduce resource wastage, improve profitability, and adapt to changing climatic conditions.

1.6 Problem description:

In today's growing world where agriculture is one of the major developing sectors of economy identifying which crop to grow has been a main problem for farmers. This project aims to present a solution to the farmers using current advanced technology like machine learning. The goal of this project is to develop a model or framework that can recommend suitable crops based on input parameters like pH and temperature values of soil.

Agriculture is a vital sector for food production and the economy, but it faces numerous challenges, including changing climate conditions, inefficient use of resources, and unpredictable market dynamics. A critical decision for farmers is determining which crops to plant in a given season, based on various factors like soil quality, climate, water availability, and market demand. Choosing the right crops ensures better yields, optimal use of resources, and improved economic returns.

However, traditional methods of crop selection often rely on historical knowledge error, which can result in suboptimal decisions. With the advent of data science and machine learning (ML), there is an opportunity to create a more precise, data-driven approach to crop recommendation. The goal of a crop recommendation system is to suggest the most suitable crops for a specific region, considering local conditions and external factors, ultimately maximizing yield, minimizing resource wastage, and improving profitability for farmers.

1.7 Objectives:

1. To design and implement machine learning algorithm to recommend which crop to grow based on soil pH and temperature.

2. Performance analysis of the two algorithms using machine learning.

3. To design and develop the user interface.

1.8 Organization of the Report:

The next chapter 2 includes Literature Survey of 5 papers followed by chapter 3 which includes system design followed by chapter 4 which has snapshots of results and descriptions about those results and the last chapter is conclusions along with future enhancements and references.

Chapter 2

LITERATURE SURVEY

Literature survey is the important part of report as it gives a direction in research. It helps to set a goal for analysis-thus giving problem statement. It is also a systematic and thorough search of all types of published literature as well as other sources including dissertation. Following are some of the research papers related to this project.

[1] AI – Assisted Farming for Crop Recommendation & Virtual Assistance Application

Author: DR. Soma Prathibha, Hariharan SV, Vignesh Kumar K, Shamini G, Kaajal Krishnamurthy, Maheshwari Avudaiappan

Year:2022

Description:

It is a fact that agriculture in India contributes significantly to employment and economic growth. The common problem Indian farmers experience is choosing the right crop for their soil, which reduces productivity. The farmers can resolve this problem through precision agriculture. Precision agriculture is characterized by a soil database collected on the farm, crop information provided by agricultural experts, and soil parameters from soil testing labs. As soon as the soil testing lab provides data to the recommendation system, the system will utilize the information and create an ensemble model. For the recommendation, we need a few parameters like NPK (nitrogen, phosphorus, potassium) ratios, and PH values. This project gives full assistance to the farmers virtually. By doing so we can prevent the suicide rates of farmers across the country. This paper concentrates specifically about the prediction of crops. The collected data about the plants and their NPK values are fed into ML models. The preferred model for crop prediction is Random Forest[1].

Selection of crops shows an essential part in the economic evolution of a developing nation like India, but selection of crops gives a vigorous part in agricultural land. Currently, the share of the agricultural sector in GDP is 19.9% in 2020-21[1]. While the suicide rate of farmers has been increasing steadily, according to the authors conducted a brief analysis of farmer suicide rates, concluding that approximately 11.2% of all suicides are farmer suicides, and provided software to connect them with the government. Amidst all the software and solutions available till date, they couldn't solve the problem proposed. The crop yield depends on multiple factors like temperature, soil conditions, and organic and financial elements. It is difficult for a farmer to decide what crop should be grown on their field to gain profit. As a result, farmers do not gain their required profit[1]. The Random Forest Algorithm is used in the suggested system. Popular machine learning algorithm Random Forest is a part of the supervised mastery approach. It can be applied to regression and class problems in machine learning. It is based on the concept of ensemble mastery, which is a method of combining different classifiers to tackle a complex problem and enhance the version's overall performance. According to the call, "Random Wooded Area is a classifier that carries several selection criteria on multiple subsets of the supplied dataset and takes the average to increase the predictive accuracy of that dataset"[1]. Instead of relying solely on one selection tree, the random forest uses predictions from all of the trees and is purely reliant on publicly available data. Farmers are unaware of what crop to grow, what is the right time to start, when to irrigate, how much water the plant requires, whether that crop can be grown on that soil and many more. The solution to all these issues is a complete application for virtual farmers to assist the farmers in their cultivation.

Advantages:

- Diverse Insights: It includes multiple methodologies, such as machine learning models and smartphone applications, to provide a comprehensive view of existing solutions.
- Identification of Gaps: Highlights limitations in existing systems, such as data inaccuracy and lack of accessibility for farmers, which can guide further research.
- Foundation for Improvement: Offers a basis to develop enhanced systems by learning from prior approaches, such as integrating real-time data and ensuring end-to-end assistance for farmers.
- Coverage of Key Parameters: Addresses essential factors like rainfall, temperature, soil type, and nutrient levels, providing a thorough analysis for crop prediction.

Disadvantages:

- Limited Data Sources: Cites datasets from sources like government websites and Kaggle, which may not always be precise or tailored to specific local conditions.
- Incomplete Solutions: Existing systems discussed are primarily limited to crop yield prediction and do not provide holistic farming assistance.

- Technological Barriers: Assumes availability of technology like smartphones and GPS, which may not be accessible to all farmers, particularly in rural areas.
- Lack of Generalization: The models are often region-specific, with some trained only for Tamil Nadu districts, limiting scalability and applicability in diverse agricultural contexts.

[2] A Decision Support Tool for District Level Planning of Agricultural Crops for Maximizing Profits of Farmers

Author: Mayank Ratan Bhardwaj, Abhishek Chaudhry, Inavamsi Enaganti, Kartik Sagar and Y. Narahari

Year:2023

Description:

A mismatch between the crops produced by farmers and the respective market demands leads to large-scale crop dumping[2]. Year on year, this leads to wastage as well as huge financial losses for the farmers. To alleviate this problem, we address the macro-level problem of district level agricultural crop planning. Our interest is in how the Government or any state administration could make an informed recommendation on which crop acreages (number of acres cultivated under each crop in each district) to grow in which districts or geospatial regions in a given state or country, so as to match the demand for the crops and maximize the profits of the farmers. In this paper, we develop a tool CROP-S (CROp Planning System) to determine an assignment of crop acreages to districts so as to maximize the profits of the farmers while simultaneously ensuring required crop security levels for each district[2]. The tool uses data about predicted demands, transportation costs, compliance ratios, and historical data about yields and prices to arrive at a nearly optimal assignment of crop acreages to districts. CROP-S uses a methodology based on genetic algorithms. We believe CROP-S will provide an effective decision support tool for the Government to issue crop recommendations to the district administrations, who in turn issue advisories to farmers. To demonstrate the effectiveness of CROP-S, we have considered the problem of allocating crop acreages to the 30 districts of Karnataka state in India[2]. We have used real world data on the top 7 crops grown in Karnataka. We show that assigning crop acreages among the various districts, as recommended by CROP-S, leads to a significant increase in the overall profits of the farmers.

This paper is motivated by the uncertainties and difficulties faced by small and marginal farmers in the world in general and in India in particular. India is the second largest producer of farm products, and approximately 50% of the Indian workforce is involved in agriculture. Agriculture in India accounts for 17-18% of the gross domestic product of the nation. India ranks first in the world based on the net cultivated area. This shows the scale at which the agricultural sector operates in India. Minor changes in crop patterns and their supply chain can lead to a huge impact on the revenues or profits of the farmers. Farmers in India struggle to sell their produce at profitable rates. This is usually caused by a mismatch between supply and demand and lack of access to markets, transportation. Lack of knowledge about other profitable alternatives is another major cause behind farmers growing nonprofitable crops. It would therefore be very useful if the farmers could be informed beforehand about which crops to produce and in what quantity, after considering various factors. Such factors include, but are not limited to, supply and demand matching, environmental conditions, transportation costs, and the presence of agriculture industries, processing centres, warehouse locations, cold storage facilities, etc. Based on the factors mentioned above, if the government can systematically carry out district level planning to determine which crops should be grown in what quantities, and in which district, it would be beneficial for the state as well as the individual farmers. In addition to the monetary profits that accrue to the farmers, there are other tangible and intangible benefits that need to be considered, such as growing crops for self consumption, bolstering bio-diversity, improving the farm's soil condition, etc. In this paper, we develop a tool CROP-S (Crop Planning System) to determine an assignment of crop acreages (number of acres cultivated under each crop in each district) so as to maximize the profits of the farmers. The tool CROP-S uses data about predicted demands, transportation costs, compliance ratios, and historical data about yield and prices to arrive at a nearly optimal assignment of crop acreages. CROP-S uses a methodology based on genetic algorithms[2]. We believe CROP-S will provide an effective decision support tool to the Government to issue crop recommendations to the district administrations, who in turn issue advisories and incentives to farmers. To demonstrate the effectiveness of CROP-S, we have considered the problem of allocating crop acreages to the 30 districts of Karnataka state in India. We have used real-world data on the top 7 crops grown in Karnataka. We show that reassigning crop acreages among the various districts, as recommended by CROP-S, leads to a significant increase in the overall profits of the farmers.

Advantages:

- Comprehensive Coverage: The survey considers a wide range of crop planning studies, including various levels of granularity (farm-level, district-level, state-level, and intercountry).
- Inclusion of Relevant Factors: It highlights critical factors influencing crop planning such as supply-demand matching, water sustainability, transportation costs, and compliance ratios.
- Novelty and Gaps: Identifies gaps in previous studies, such as the lack of integration of postproduction factors like transportation costs and demand variability, and introduces innovative concepts like compliance ratios.
- Contextual Relevance: Focuses on crop planning challenges specific to Karnataka, India, which makes the study locally relevant and actionable.
- Analytical Depth: The survey provides insights into the limitations of static optimization approaches and promotes dynamic, data-driven methodologies.
- Technological Insight: Emphasizes genetic algorithms for optimization, showcasing modern and effective computational techniques.

Disadvantages:

- Limited Scope of Post-Production Factors: While it identifies some factors, others like warehouse capacities, processing units, and export-import data are not deeply explored.
- Geographical Bias: Focused primarily on India, particularly Karnataka, it may not fully generalize to other regions or international contexts.
- Comparative Analysis Deficit: The survey could benefit from a more detailed comparison between different optimization techniques (e.g., genetic algorithms vs. simulated annealing).
- Historical Data Dependency: Heavy reliance on historical yield and price data may limit adaptability to future climate or market changes.
- Narrow Crop Selection: The analysis is restricted to the top seven crops in Karnataka, potentially neglecting less prominent crops that might have economic significance.
- Assumptions: Simplifying assumptions, like Karnataka as a "closed box" for crop consumption, may overlook broader market dynamics.

[3] An Efficient Machine Learning Technique for Crop Yield Prediction

Author: Niveditha Singh, Sachin Patel, Shubha Dubey

Year:2022

Description:

The field of agricultural research is flourishing, and the difficulties that lie ahead are being overcome with the aid of technological development. It has been determined that this is very helpful for the economic development and progress of any country. Agricultural growth to be improves by incorporating technology[3]. In order to make strategic plans in agricultural commodities for import export policies and to enhance farmer earnings, an early and accurate assessment of crop achievement is important in quantitative and financial evaluation at the field level. One of the tough problems in agriculture is predicting crop yields with the use of machine learning algorithms in order to forecast a larger crop output. Using this work, provide a powerful approach of machine learning to forecast agricultural yields[3]. By using machine learning technique can be helpful for predicting the crop yield. An experimental analysis shows that using machine learning technique crop yield productivity can be enhanced.

Agriculture is the backbone of the economy, more than half population depends on agriculture for their livelihood, making it an essential sector of the country's economy. Machine learning methods are used to forecast agricultural yields in light of variables including precipitation, crop type, and weather[3]. Random Forest, the most widely used and potent supervised machine learning algorithm, is capable of both classification and regression. In crop selection, they help minimize losses in output yield in the face of environmental disturbances. Long-term agricultural sustainability has been seriously threatened by weather, climate, and associated environmental variables. Significantly, machine learning (ML) provides a decision-support tool for Crop Yield Prediction (CYP), which can be used to determine things like which crops to produce and how to care for them when they are in season.

The primary goal of crop yield estimation is to increase agricultural crop output, and it achieves this via the use of a number of tried-and-true models. Due to its efficacy in many fields, including forecasting, defect detection, pattern recognition, and so on, machine learning is seeing widespread adoption throughout the globe[3]. Predicting harvest yields is a major issue in agriculture. The findings of this research will help farmers predict their crop's yield before it's even planted in the field. Properly timed guidance to predict future crop production and analysis is necessary to help farmers maximize agricultural produce. The current situation of agricultural production throughout the globe is presented first, followed by a short introduction to widely used characteristics and forecasting methodologies. The difficulty of selecting seeds and forecasting yields is compounded by the vast datasets that farmers must sift through. This is a major problem in agriculture. Scientists utilize a wide range of data mining techniques to boost agricultural development and find solutions to industry difficulties. Diseases in crops, and the losses they cause, may be predicted using a variety of data mining approaches, including classification and prediction. Bacterial, fungal, and viral pathogens are all possibilities. Plants are susceptible to a wide variety of illnesses, including bacterial wilt, black knot, curly top, etc. Many different kinds of insects and microorganisms are at blame for these illnesses. Various classifiers, including Decision Trees, Neural Networks, Naive Bayes Classifier, Random Forests, Support Vector Machine, and K-Nearest Neighbour, have been employed in an analysis of crop disease prediction using Machine Learning[3]. The weaker classifiers' performance may be boosted by using an ensemble model.The loss of production is typically the result of crop diseases. Unfortunately, rural farmers in underdeveloped nations often lack access to cutting-edge tools for early identification of crop diseases and subsequent remedial measures. In recent years, deep learning methods have emerged as the most efficient means of analysing agricultural data and producing reliable outcomes.

Forecasting accurately and making timely estimates is essential, yet it is a difficult task due to many interrelated factors. Wheat, peas, rice, pulses, tea, sugar cane, greenhouses, cotton, soybeans, and maize are only a few of examples of crops that can be used to predict crop yields. Extremely large data sets and heightened vigilance are essential in the agricultural sector[3]. Current high throughput grain counting technologies have various drawbacks, the most prominent of which is their inability to distinguish and separate the overlapping grains and panicles that are characteristic of rice farms. By identifying and counting the total number of rice panicles and estimating the total quantity of grains, propose and discuss a unique approach to calculate the total rice production for a dataset including photos of drought-affected and managed rice farms.

Advantages:

- Prediction: Broad Coverage of Techniques: A well-executed literature survey encompasses a variety of machine learning (ML) approaches like regression models, neural networks, and ensemble methods, providing a comprehensive overview.
- Identification of Strengths and Weaknesses: Highlights the relative performance of different ML models, helping to identify the most effective techniques for specific scenarios (e.g., deep learning for large datasets or regression for simple datasets).
- Incorporation of Diverse Datasets: Often, such surveys examine methods tested on various datasets, making the findings relevant across different climates, soil types, and crops.
- Comparison with Traditional Methods: The survey typically compares ML techniques with conventional statistical or rule-based models, showcasing improvements in accuracy and efficiency.
- Trend Analysis: Provides insights into emerging trends like the use of big data, remote sensing, IoT, and cloud computing for crop yield predictions.
- Scalability Potential: Highlights ML techniques' ability to scale for regional or global applications, especially when integrated with geospatial and temporal data.
- Focus on Key Metrics: Surveys generally focus on critical evaluation metrics (e.g., RMSE, MAE, accuracy) to assess model effectiveness.
- Practical Applications: Offers insights into how ML can aid policymakers, farmers, and agricultural industries, bridging research and real-world applications.

Disadvantages:

- Limited Dataset Diversity: Many surveys rely heavily on datasets from specific regions or crops, making generalization to other areas or crops challenging.
- Neglect of Practical Challenges: Often focuses on model performance without addressing real-world implementation challenges like data quality, availability, and computation costs.
- Overemphasis on Accuracy: May prioritize accuracy over other crucial factors like interpretability, robustness, and computational efficiency.
- Lack of Integration Focus: Few surveys address the integration of ML techniques with existing agricultural systems or other technologies like drones or IoT sensors.

- Data Dependency: Machine learning models often require large and high-quality datasets, which might not be available for all regions or crops.
- Technological Bias: Some surveys may disproportionately highlight advanced ML techniques (e.g., deep learning) while underrepresenting simpler, equally effective methods for specific cases.
- Rapidly Evolving Field: ML technology evolves quickly, so surveys risk becoming outdated, failing to include the latest models or breakthroughs.
- Ethical and Socioeconomic Factors: Rarely discuss how reliance on ML might impact small scale farmers, data privacy, or accessibility in low-income regions.

[4] Crop Advancement with Machine Learning

Author: Dr. Rajendra Pawar, Vishnu Bisht, Ritika pandey, Urvik Lodhe, Aaryan Chouksey, Sarthak jatale, Aryan Pawar

Year:2023

Description:
Crop prediction is a crucial task in agriculture that can help farmers and agricultural researchers decide which crops to grow based on environmental variables such as soil type, temperature, rainfall, and humidity. In this project, we propose a data-driven approach for crop prediction using K-means clustering and classification algorithms[4]. We use K means clustering to group the data into different clusters based on similarities in the environmental variables, and then we use a classification algorithm such as decision trees or support vector machines to predict the best crop to grow in a particular location. This approach can help farmers and agricultural researchers make more informed decisions about crop selection and improve crop yields. We validate our model by comparing historical data on crop yields to our predictions and conducting experiments by growing different crops in different locations. Overall, this project demonstrates the potential of machine learning algorithms for crop prediction in agriculture[4]. C. Research Objective or Research Questions: The creation of a crop compatibility prediction model based on soil and environmental variables is the main goal of this research work. The research questions that this study aims to answer are: What are the key soil and environmental factors that influence crop suitability? Can k-means clustering be used to group soil types based on their nutrient content? Can logistic regression be used to predict missing values in the dataset?

Agriculture is the backbone of many economies worldwide, and crop production is crucial for food security and economic growth. However, the yield and quality of crops are significantly affected by the soil's nutrient content, temperature, humidity, and rainfall. It is essential to determine which crop is most suitable to grow in a particular soil type to ensure maximum yield and quality[4]. This problem is complex, and several factors need to be considered to make an informed decision. Recent literature indicates that there has been a growing interest in crop suitability prediction models that take into account soil and environmental factors.

The creation of a crop compatibility prediction model based on soil and environmental variables is the main goal of this research work[4]. The research questions that this study aims to answer are: What are the key soil and environmental factors that influence crop suitability? Can k-means clustering be used to group soil types based on their nutrient content? Can logistic regression be used to predict missing values in the dataset? How accurate is the proposed model in predicting the most suitable crop for a given soil type and environmental conditions?

Advantages:

- Comprehensive Overview: Summarizes diverse ML techniques used in crop advancement, such as disease detection, yield prediction, irrigation optimization, and soil quality analysis.
- Interdisciplinary Approach: Integrates insights from multiple fields, including agriculture, computer science, and environmental science, for a holistic perspective.
- Highlights Efficiency Gains: Demonstrates how ML reduces labour, time, and resource usage by automating and optimizing traditional agricultural practices.
- Real-World Applications: Covers practical applications like precision agriculture, pest management, and climate adaptation strategies, making findings relevant to stakeholders.
- Scalable Solutions: Showcases how ML techniques can be scaled to regional, national, or global levels, aiding policy and decision-making processes.
- Addresses Big Data Challenges: Explores the use of satellite imagery, IoT data, and weather forecasts, leveraging big data for enhanced agricultural insights.
- Trend Analysis: Identifies emerging trends such as the integration of ML with robotics, remote sensing, and drone technology.

- Quantifiable Impact: Often highlights improvements in yield, cost savings, or resource efficiency through case studies and performance metrics.
- Encourages Innovation: Inspires further research by identifying gaps in existing literature, such as the need for more region-specific models or the inclusion of underrepresented crops.
- Environmentally Friendly Solutions: Focuses on ML methods that promote sustainable farming, such as reducing fertilizer overuse or optimizing water consumption.

Disadvantages:

- Data Dependency: Relies heavily on the availability of large, high-quality datasets, which may not exist for all regions or crop types.
- Overemphasis on Advanced Models: Advanced ML techniques like deep learning are often prioritized, even when simpler models could be more practical or interpretable for specific tasks.
- Generalization Challenges: Findings may not generalize well across different geographical locations, climates, or farming practices.
- Implementation Gaps: Focuses on theoretical advancements without addressing real world implementation challenges, such as the cost of technology or the digital divide.
- Neglect of Socioeconomic Factors: Limited discussion on how small-scale or resource poor farmers can adopt these technologies.
- Lack of Comparative Insights: Few surveys rigorously compare ML approaches to alternative methods, such as traditional agricultural techniques or other technological solutions.
- Ethical and Privacy Concerns: Rarely considers the ethical implications of data collection, farmer privacy, or potential biases in predictive models.
- Technological Obsolescence: With rapid advancements in ML, findings from older surveys risk becoming outdated quickly.
- Environmental Impacts: Limited exploration of the energy consumption and carbon footprint associated with large-scale ML implementations in agriculture.
- Insufficient Focus on Integration: Does not always explore how ML tools can seamlessly integrate into existing agricultural systems and workflows.

[5] Crop Recommendation using Machine Learning

Author: 1st Ramachandra A C, 2nd Venta Ankitha, 3rd Idupulapati Divya, 4th Parimi Vandana, 5th H S Jagadeesh

Year:2021

Description:

One of the key economic drivers in India is agriculture. Farmers now cultivate crops using lessons learned from the previous century[5]. One of the most crucial elements of farm planning is crop selection. Losses are reduced when farmers are well-informed on the crops that will thrive in their soil and climate. Many factors affect crop yield, including specific meteorological conditions and soil characteristics (such as soil N, P, K values, soil moisture, etc.) [5]. Various datasets including these traits were gathered and examined. The data analysis process, which evaluates each component of the data using a variety of analyses and logical reasoning, is crucial. Agricultural monitoring and the food business use a variety of models thanks to the development of machine learning algorithms. Post-season data on crop types, however, won't help with crop estimation and monitoring during the growing season[5]. On the other hand, by mapping pre-season crops, early warnings of agricultural production and supply chains can be provided, lowering trade tensions and agricultural hazards. This work analyses and suggests pre-season crop-type maps using a variety of learning algorithms based on past crop-type data. As examples of machine learning algorithms, Support Vector Machines, Random Forest, and Naive Bayes are utilized for computation, the accuracy is calculated and compared. The crop prediction also checked based on the parameters considered for computation.

The most significant economic sector in terms of the population is agriculture, which also has a significant impact on India's overall socioeconomic structure. The choice of crops is a crucial component of agricultural planning[5]. The need for agricultural production will rise over time. Agriculture's economic contribution to India's GDP has continuously decreased along with the country's economic expansion because of its industrialization revolution. Using technologies to get the appropriate output is the key issue the Indian agricultural sector is currently experiencing. Farmers may not be familiar with many crop forecasting models due to their complexity or cost effectiveness. Consequently, it is necessary to create a model that is straightforward, user friendly, affordable and provides the necessary precision. Many scholars have used machine learning techniques in the past to increase the nation's agricultural growth.

To predict crop yields, consider a variety of factors like rainfall, temperature, and seasons. No additional machine learning techniques were used on the dataset. Lack of comparison, quantization, and other algorithms prevent the provision of a suitable algorithm[5]. In turn, this offers the greatest crops to grow considering the existing environmental circumstances. The suggested system considers the type of land the farmer owns as well as rainfall in the past, present, and future. Machine learning algorithms are employed to forecast which crops are most suited to the current circumstances, resulting in more accurate predictions[5]. The dataset was constructed using the following arguments: soil values for N, P, and K, rainfall, temperature, and Ph.

Based on the previously mentioned data fields such as nitrogen, phosphorous, potassium, ph, humidity, temperature, rainfall, the crop is predicted using machine learning techniques like Support Vector Machine (SVM), Random Forest, and Naive Bayes[5]. The variety of crops that are predicted: Rice, Maize, chickpea, kidney beans, pigeon peas, moth beans, mung beans, muskmelon, apple, orange, black gram, lentil, pomegranate, banana, mango, cotton, jute, grapes etc.

Advantages:

- Comprehensive Understanding:

Provides a clear overview of existing techniques such as Decision Trees, Random Forests, Support Vector Machines (SVM), Neural Networks, etc.

Helps identify the best practices and state-of-the-art algorithms for crop prediction.

- Problem Identification:

Highlights challenges in the domain, such as data collection, preprocessing, and model scalability, which can guide future research.

- Evaluation of Models:

Analyses the performance of different models in terms of accuracy, precision, and other metrics, allowing a comparison to choose the most suitable approach.

- Guidance for Future Work:

Suggests trends and areas for innovation, like incorporating real-time weather data or precision agriculture techniques.

- Resource Optimization:

Identifies the most efficient algorithms and methodologies, which can save computational and financial resources in real-world implementations.

Disadvantages:

- Limited Dataset Information:

  While the paper mentions the data used, details about dataset size, source authentically, and preprocessing methods are insufficiently explained, which might affect reproducibility.
- Algorithm Limitations:

  Although multiple algorithms are utilized, the reasons behind the choice of these algorithms or why others were not considered (e.g. deep learning techniques) are not discussed in detail.
- Feature Importance Bias:

  The paper highlights feature importance but shows a strong reliance on humidity and rainfall, which may not generalize well to regions with different climatic patterns.
- Regional Constraints:

  The dataset and models are likely specific to Indian conditions, limiting global applicability without significant modifications.
- Real-World Integration Challenges:

  While the model claims good accuracy, factors like data availability in real-time, model interpretability and farmer adaptability to technology are not fully addressed.
- Scalability Issues:

  There is no in-depth discussion on how the model handles large-scale data, different crops or variable environmental conditions over time.

Chapter 3

SYSTEM DESIGN

System design for machine learning refers to the process of designing the architecture and infrastructure necessary to support the development and deployment of machine learning models. It involves designing the overall system that incorporates data collection, preprocessing, model training, evaluation, and inference. System design in machine learning (ML) involves architecting end-to-end solutions that can efficiently handle tasks such as data preprocessing, model training, deployment, and monitoring. Below is an overview of key aspects in the system design of an ML system.

3.1 Block diagram for identifying the crop using soil parameters

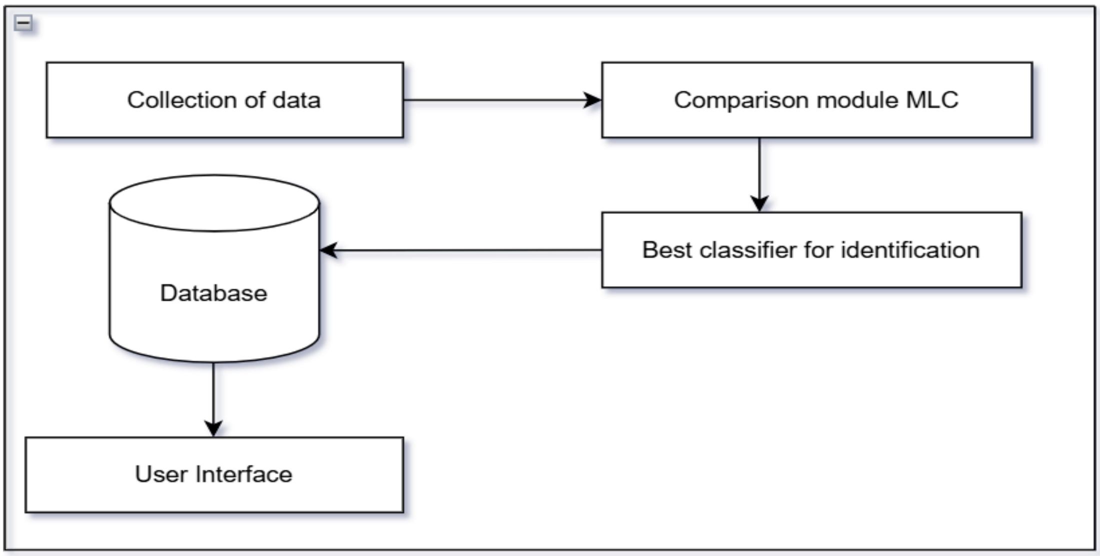

Fig: 3.1 Block diagram of crop recommendation system using ML

This Fig 3.1 shows the block diagram of crop recommendation system

- Collection of data set from the Kaggle website.
- Comparison module machine learning classifier (MLC)

  Random forest - Random Forest algorithm is a powerful tree learning technique in Machine Learning. It works by creating a number of Decision Trees during the training phase. Each tree is constructed using a random subset of the data set to measure a random subset of features in each partition. This randomness introduces variability among individual trees, reducing the risk of overfitting and improving overall prediction performance. Random forest generates many decision trees, each trained with a random subset of the training data. The algorithm then combines the predictions from each tree to produce a final result. We can see this in above Fig 3.1 Random forests or random decision forests are an ensemble learning method for classification, regression and other tasks, that operate by constructing a multitude of decision trees at training time and outputting the class that is the mode of the classes (classification) or mean prediction (regression) of the individual tree.

Support Vector Machine (SVM) - A Support Vector Machine (SVM) is a powerful machine learning algorithm widely used for both linear and nonlinear classification, as well as regression and outlier detection tasks. These are highly adaptable, making them suitable for various applications such as text classification, image classification, spam detection, handwriting identification, gene expression analysis, face detection, and anomaly detection. SVM algorithm is a supervised machine learning technique that uses a hyperplane to classify data or solve regression problems:

How it works: The SVM algorithm finds the optimal hyperplane that maximizes the margin between the closest data points of different classes. The margin is the distance between the hyperplane and the nearest training examples.

- Best classifier for identification

  Comparing these two algorithms, Random Forest came up with 75% accuracy and SVM came up with 60% accuracy, based on this, random forest is chosen as the best algorithm which is implemented in the prediction.

- Data base connectivity

  Data base connection is done through Flask server.

- User Interface

  User Interface like login page is created using HTML and CSS file in VS code and then data base is connected with user interface.

3.2 Flow chart of crop recommendation system using MLC

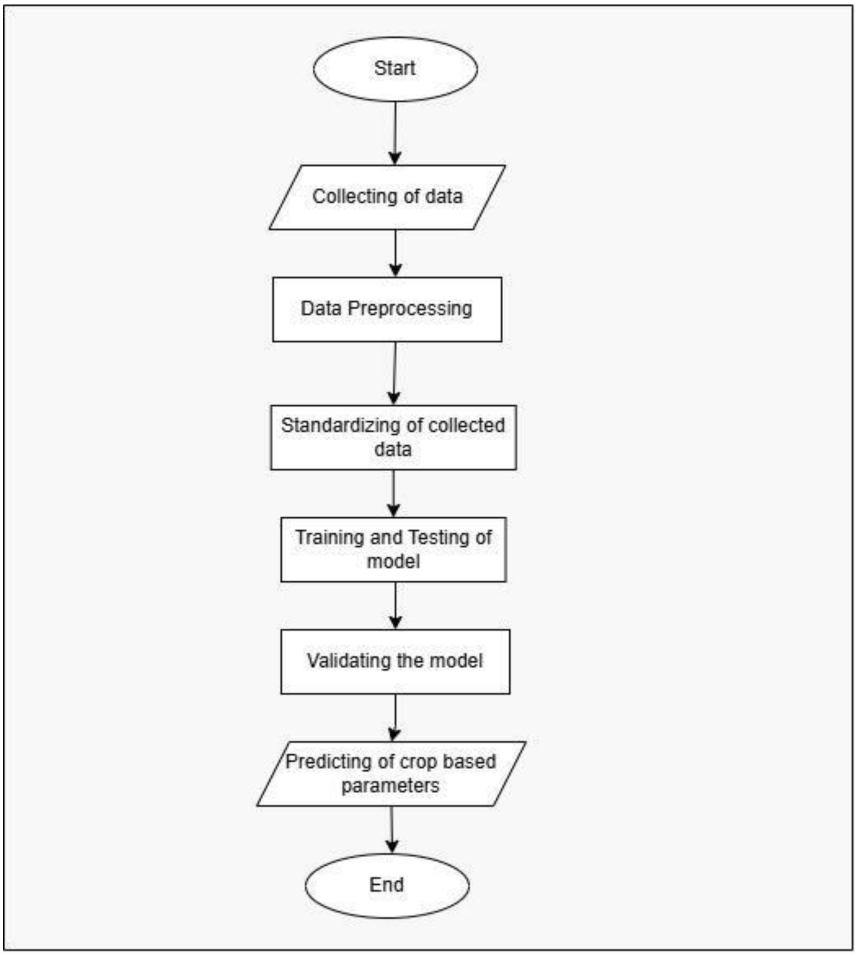

Fig:3.2 Flow Chart of Crop Recommendation System

This Fig 3.2 shows flow chart of crop recommendation system

- **Collection of data set:** Collection of data set is from the Kaggle website
- **Data Preprocessing**: It ensures the data is clean, consistent, and ready for model training.

  The steps involved in data preprocessing are:

  1. Understand the dataset
  2. Data Cleaning
  3. Feature Engineering
  4. Handle Class Imbalance
  5. Feature Selection
  6. Split the dataset
  7. Outlier Detection and handling

8. Data Augmentation
9. Pipeline Augmentation

- **Standardizing of collected data:** It ensures that all features have comparable scales, which helps many machine learning models (e.g. SVM, Random Forest, K-NN, Logistic regression) perform better.
- **Training and testing of the model:** It is a model with a dataset that contains features like pH, temperature, humidity, rainfall.
- **Validating the model:** It is a crucial step in ensuring the reliability and generalizing of a machine learning model in a crop recommendation system.
- **Predicting of the crop, based pH and temperature:** Comparing the two algorithms, i.e. Random Forest came up with accuracy about 75% and SVM came up with 60% accuracy. Based on this, random forest is chosen as the best algorithm which is implemented in the prediction.
- **Data base connectivity:** Database connectivity is done through Flask server.
- **User Interface:** User interface like login page is created using HTML and CSS file in VS code and then data base is connected with user interface.

3.3 Analysis table to predict the crop

Table 3.1 Analysis of table using soil parameters such as pH, temperature and humidity to predict the crop

pH	Temperature	Humidity	Predicted Crop
6.5-7.0	**34**	**88**	**Rice**
6.5-7.0	**12-25**	**80**	**Wheat**
5.5-7.5	**18-27**	**89**	**Maize**
5.5-7.0	**20-27**	**75**	**Tomato**
6.0-7.5	**21-27**	**90**	**Sugarcane**

By analysing the inputs of the soil parameters which is given in the analysis table, is used to predict the crop. This table simplifies decision making, allowing farmers to focus on other farm activities. It reduces unnecessary use of fertilizers and amendments. By planting crops which is well suited to soil conditions, farmers maximize yields. It also increases profits by aligning crops with market demand and soil suitability.

Chapter 4

RESULTS AND SNAPSHOTS

This chapter includes fix sections. First section consists of snapshots of the login page of crop recommendation system. Second section consists of parameters which is used to predict the crop. Third section is about entering the suitable values for the parameters which is used to predict the respective crop. Fourth section includes the graph of accuracy vs soil features using Random Forest algorithm. Fifth section consists of prediction of crop with its accuracy. The final section is of prediction of crop with accuracy vs soil features of both Random Forest and SVM algorithms.

4.1 Login Page of crop recommendation system

Login Page for a crop recommendation system is a user authentication interface that allows registered users to securely access the system. This system provides crop suggestions based on input factors such as soil type, weather, and other environmental conditions. It protects sensitive data and recommendations and ensures that only authorized users can access the system. This page also serves as the entry point to ensure the users interact with the system efficiently and securely

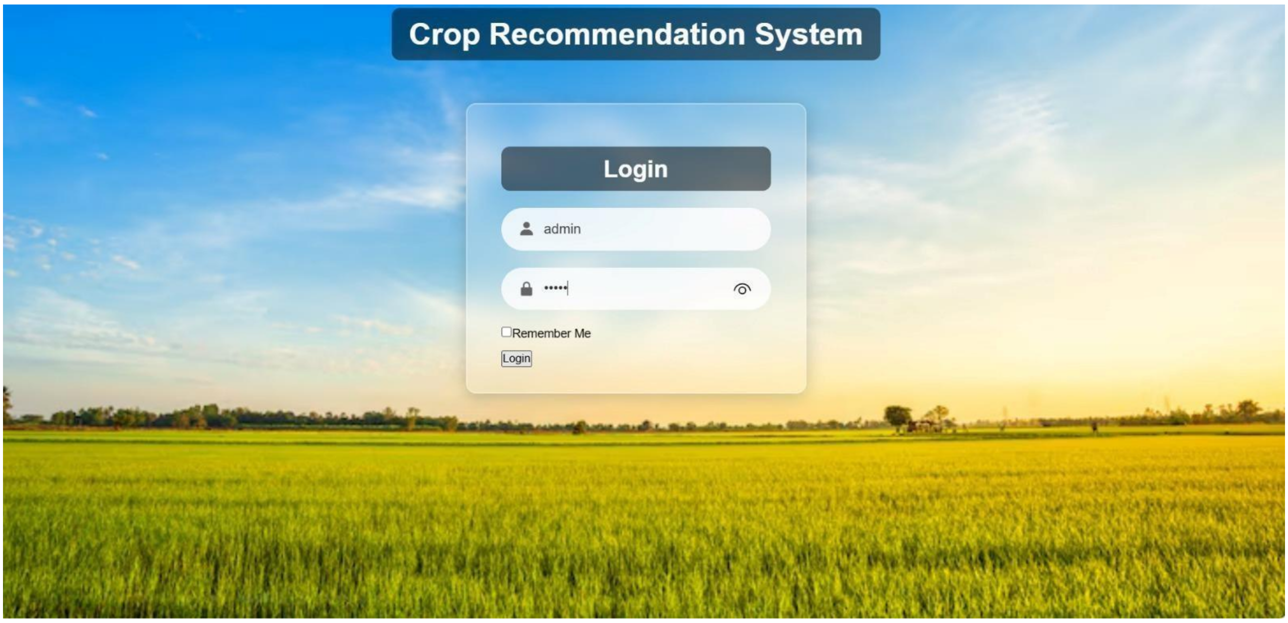

Fig:4.1 Login Page of crop recommendation system

The Fig 4.1 shows created user interface or a login page which can be sent to the farmers in various locations across the world, in which they can login the page and fill the information and it gets login to the page.

4.2 Predicting crop using parameters

This page involves utilizing various input factors to recommend the most suitable crops for a particular region or condition. This process can use statistical models, machine learning algorithms or expert systems to analyse data and make recommendations. It can include the parameters

- Soil parameters which consist of soil type (e.g. sandy, clay, loamy), soil pH, nutrient content (Nitrogen, Phosphorous, Potassium, etc)
- Climate parameters which may include temperature (minimum and maximum), rainfall (annual or seasonal), humidity levels, sunlight
- Geographical parameters which consist of location or region, altitude
- Agricultural parameters may include irrigation and crop rotation history
- Economic factors like market demand, cost of cultivation, profitability of crops etc

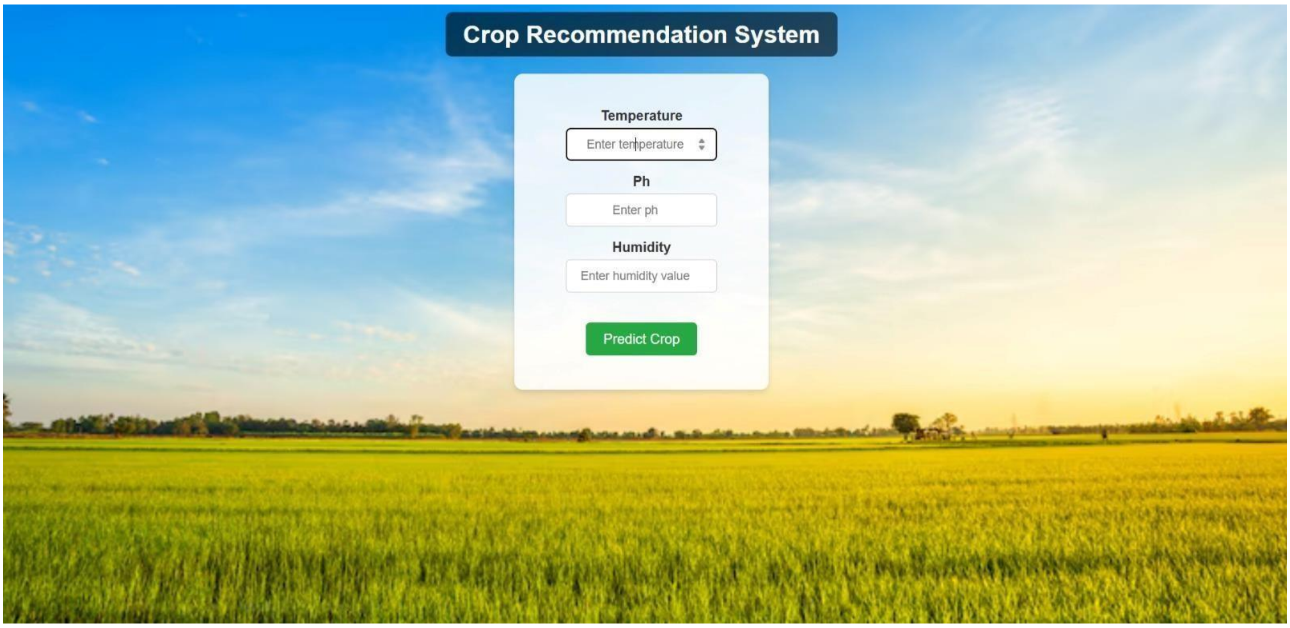

Fig:4.2 Predicting crop using parameters

Fig 4.2 shows after entering the login page using some of the parameters like pH, temperature and humidity, crops can be predicted based on soil features.

4.3 Entering of required parameters

The crop prediction parameters page allows users to input essential data to predict the most suitable crop for their region or field. This page serves as an interface for data collection, ensuring all relevant parameters are provided for accurate predictions. This page is user friendly forms to enter key parameters. It has dropdowns, sliders or text boxes for various inputs. It ensures users provide valid and complete data. For example, ensuring soil pH is between 0-14, temperature is realistic etc. It includes pre-filling values based on location (e.g. weather API integration) and conditional inputs based on selected soil type or region. It consists of "Predict Crop" button to process inputs and display recommended crops.

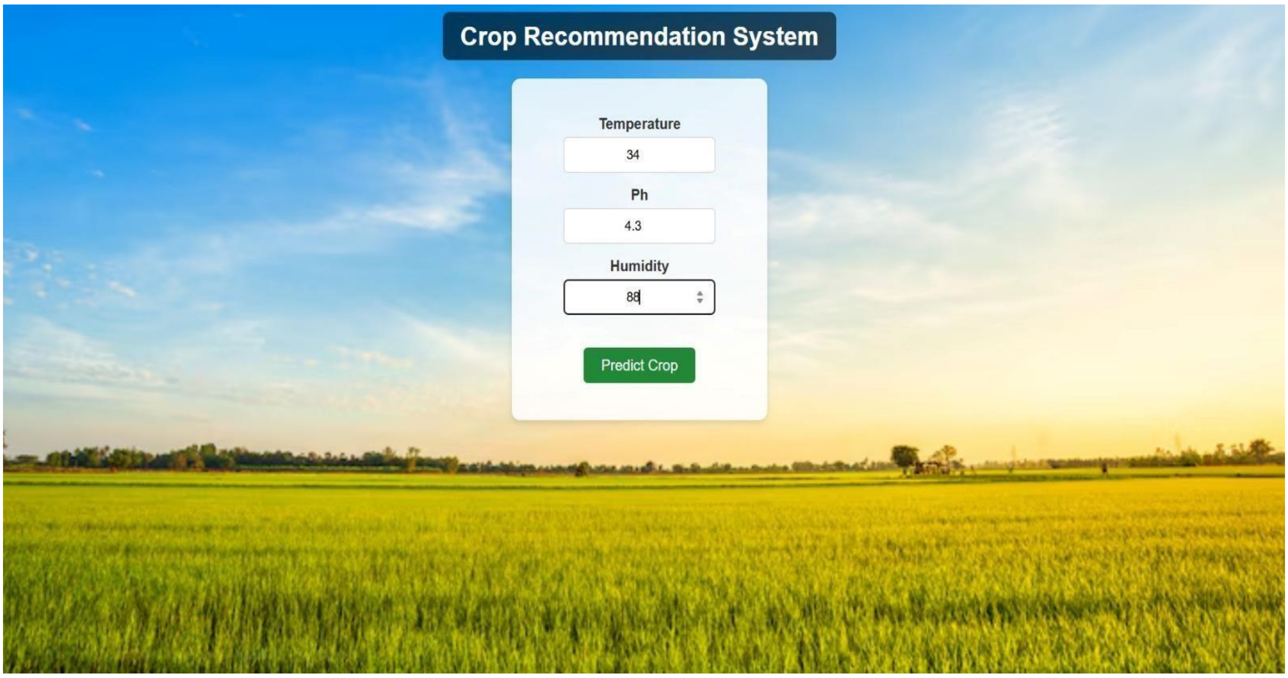

Fig: 4.3 Entering of required parameters

Fig 4.3 shows after getting into the login page, the next page consists of parameters like pH, temperature and humidity. Then inputs based on the soil features are entered to predict the crop.

4.4 Accuracy vs Soil features using Random Forest algorithm

In order to visualize the relationship between accuracy and soil features using random forest algorithm, evaluate the model's performance using metrics accuracy on validation or test data and plot the graph. This graph helps to visualize which soil features are most critical for crop prediction. Farmers can focus on monitoring and optimizing key features. It highlights essential parameters for testing and saving money of the farmers on unnecessary tests. It enables personalized crop suggestions based on influential soil features. The graph identifies features with the highest accuracy contribution. It also prevents soil degradation by selecting crops that maintain soil health.

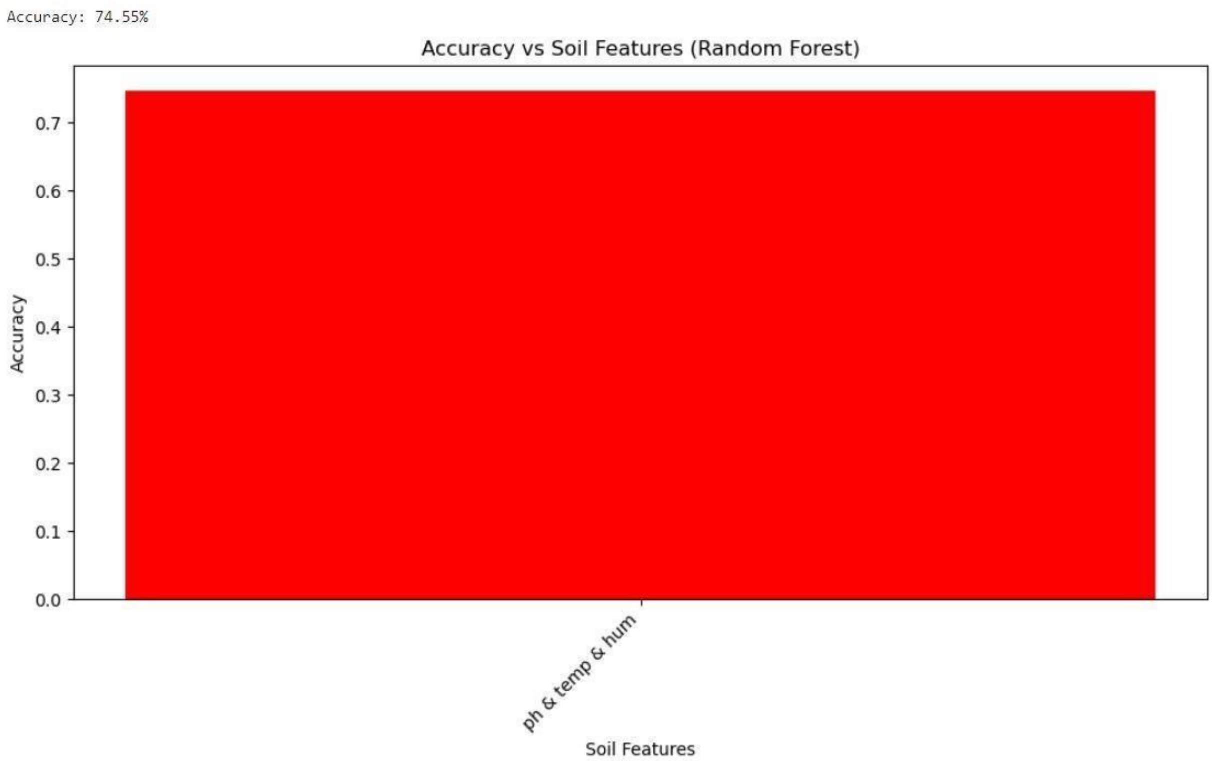

Fig:4.4 Accuracy vs Soil features using Random Forest algorithm

Fig 4.4 shows after the evaluation of the accuracy of random forest algorithm, accuracy on the y-axis vs soil features i.e. pH, temperature and humidity on the x-axis graph is plotted. It displays the contribution of each soil features to the prediction accuracy.

4.5 Accuracy vs Soil features using Random Forest and SVM algorithm

To perform crop prediction with accuracy and plot a graph showing the accuracy of the model, implementation will train a Random Forest model, predict the crop, calculate the accuracy and plot a graph comparing predicted vs actual crop labels. Crop has been predicted by entering the inputs of soil parameters and accuracy has been come up using random forest algorithm and then both the predicted crop with its accuracy is shown in a graph based on the soil features of the crop and its accuracy. The predicted crop with the graph representing the accuracy of two algorithms has been showed in result snapshot i.e. **Fig 4.5**. Here accuracy of both the algorithms - Random Forest and SVM algorithms are represented in a graph with which the name of the crop has been predicted.

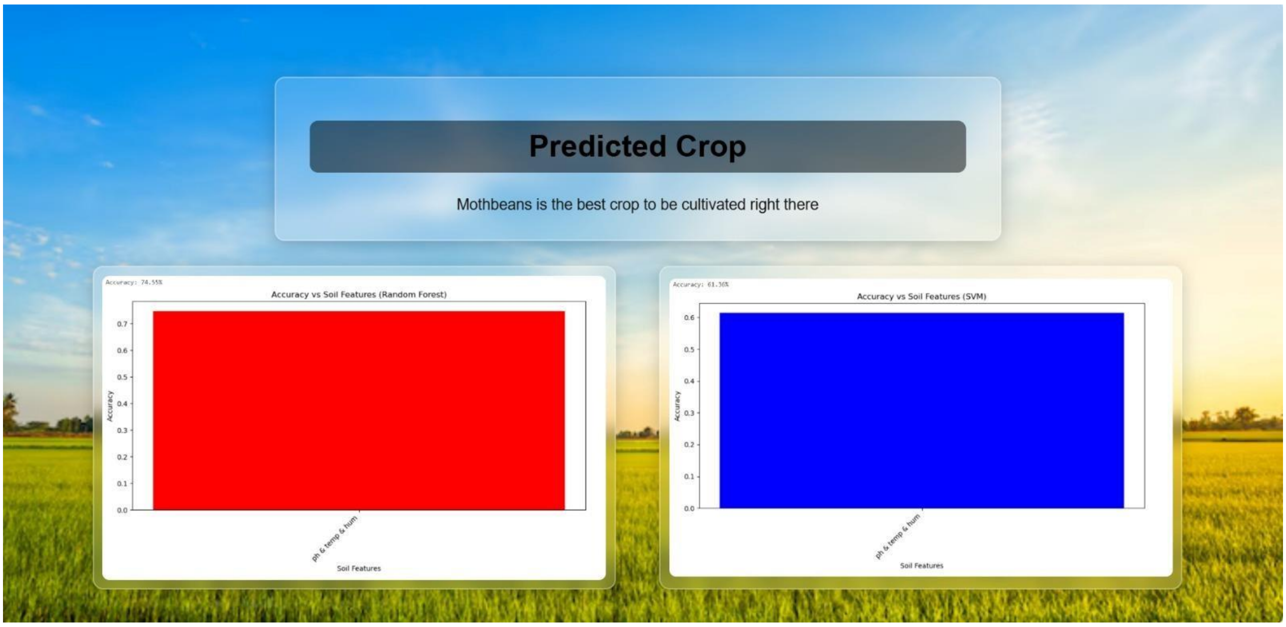

Fig: 4.5 Accuracy vs Soil features using Random Forest and SVM algorithm

Above page is result snapshot that shows the comparison of accuracy between the Random Forest and Support Vector Machine. Random Forest has come up with accuracy 75% and SVM with 60% accuracy. In order to predict the crop, accuracy of Random Forest is chosen because of its high accuracy when compared to SVM. This is implemented for the further process to predict the crop. For various inputs of soil parameters, distinct crops are predicted.

Chapter 5

CONCLUSION

The analytical process started from data cleaning and processing, missing value, exploratory analysis and finally model building and evaluation. Finally, we predict the crop using machine learning algorithm with different results. This brings some of the following insights about crop prediction. As maximum types of crops will be covered under this system, farmer may get to know about the crop which may never have been cultivated and lists out all possible crops, it helps the farmer in decision making of which crop to cultivate. Also, this system takes into consideration the past production of data which will help the farmer get insight into the demand and the cost of various crops in market.

FUTURE ENHANCEMENT

- Using neural networks, so that they may give better results.
- In this work we use parameters such as pH, humidity and temperature to predict crops along with this we can also predict which fertilizer to be used for the particular crop.
- This also involves machine learning model that can predict the crop yield based on pH, temperature, rainfall, humidity and crop type.
- The system also provides step by step guidance, from sowing to harvest, enhancing the decision making for farmers.

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